

6. DAGs

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Jaewon (“Jay-one”) Yoo

National Tsing Hua University

Where are we? Where are we going?

Up to now: Saw how to estimate the ATE with regression under selection on observables.

Now:

- How do we know if no unmeasured confounders holds? (i.e., What covariates do we need to condition on?)
 - One way, from the assumption itself: $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i \mid \mathbf{X}_i$
 - Include covariates such that, conditional on them, the treatment assignment does not depend on the potential outcomes.
 - Somewhat circular
 - Another way: use DAGs and look at back-door paths.
- What if this assumption doesn't hold?
 - Sensitivity analysis: try to vary the amount of unmeasured confounding to see if it changes the effect.

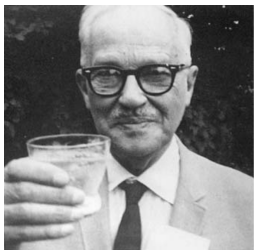


Sources: <https://www.redbubble.com/i/sticker/D-Ya-Like-Dags-by-salamincheese/27407958.EJUG5>

p.s.: from an old British-American comedy film starring Brad Pitt

1/ DAGs

Neyman-Rubin Potential Outcomes Model

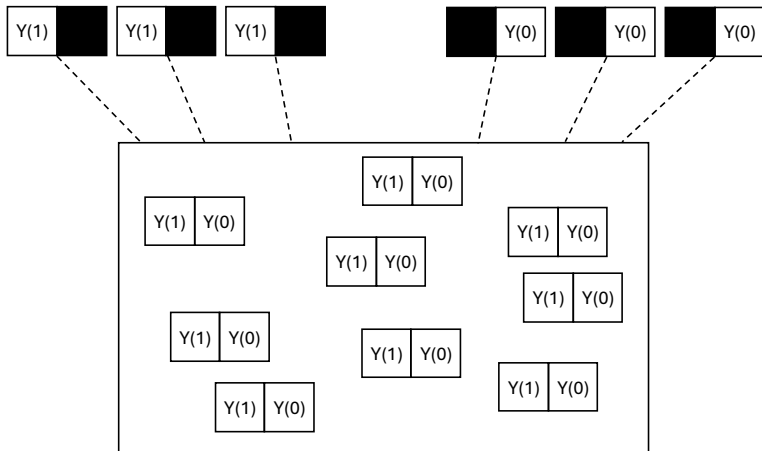


Jerzy Neyman (1894–1981)



Donald Rubin (1943–)

Neyman Urn Model



- Neyman-Rubin Potential Outcomes Model
- Defines causal effects as contrasts of POs! (Week 1)

Rubin CM: Average Treatment Effect (ATE)

Suppose we observe a population of 4 units:

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	3	3	0	3
2	1	1	1	1	0
3	0	0	1	0	1
4	0	1	1	1	0
$\mathbb{E}[Y_i(1)]$			1.5		
$\mathbb{E}[Y_i(0)]$				0.5	
$\mathbb{E}[Y_i(1) - Y_i(0)]$					1

- $\tau_{\text{ATE}} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[\tau_i] = \frac{3+0+1+0}{4} = 1.$
- Why $\tau_{\text{ATE}} \neq \hat{\tau}_{\text{diff}}$? When would they be equal?

Rubin CM: ATE on the Treated (ATT)

Again suppose we observe a population of 4 units:

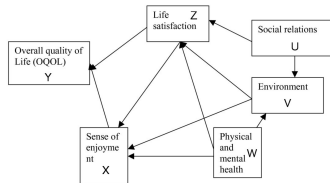
i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	3	3	0	3
2	1	1	1	1	0
3	0	0	1	0	1
4	0	1	1	1	0
$\mathbb{E}[Y_i(1) D_i = 1]$			2		
$\mathbb{E}[Y_i(0) D_i = 1]$				0.5	
$\mathbb{E}[Y_i(1) - Y_i(0) D_i = 1]$					1.5

- $\tau_{\text{ATT}} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] = \mathbb{E}[\tau_i \mid D_i = 1] = \frac{3+0}{2} = 1.5.$
- Why does $\tau_{\text{ATT}} \neq \tau_{\text{ATE}}$?

An Alternative Causal Model: Causal Graphs

- Did social scientists not do causal inference before Rubin? **No!**
- The old paradigm: **structural equation modeling** and **path analysis**

1. Postulate a causal mechanism and draw a corresponding **path diagram**



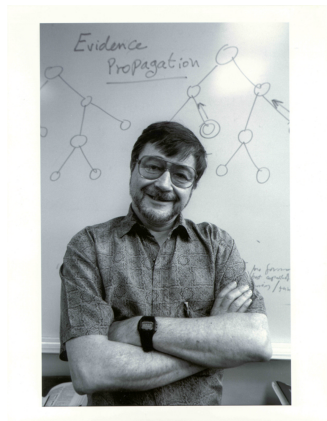
2. Translate it into a (typically linear) system of equations:

$$Y = \alpha_0 + \alpha_1 X + \alpha_2 Z + \varepsilon_\alpha$$

$$X = \beta_0 + \beta_1 Z + \beta_2 W + \beta_3 V + \varepsilon_\beta \quad \dots$$

3. Estimate α , β , etc. typically assuming normality and exogeneity
- Went out of fashion (until... Pearl's attack!):
 - Strong distributional/functional form assumptions
 - No language to distinguish causation from association

Pearl's Attack



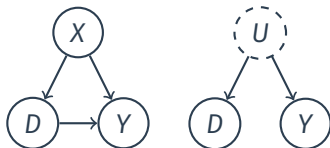
Judea Pearl (1936–) proposed a new causal inference framework based on nonparametric structural equation modeling (NPSEM)

- Originally a computer scientist
- Previous important work on artificial intelligence
- *Causality* (2000, Cambridge UP)
- Won the Turing Award in 2011 for his causal work

Pearl's framework builds on SEMs and revives it as a formal language of causality.

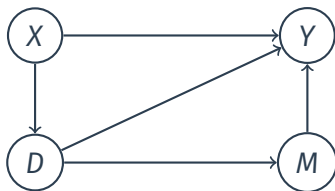
Directed Acyclic Graphs

- **Directed acyclic graphs** (DAGs) describe the causal structure of variables



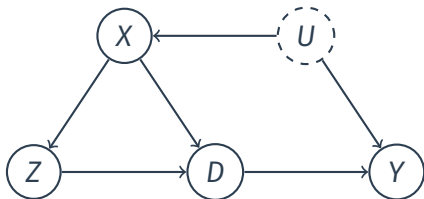
- **Nodes/vertices:** observed (solid) or unobserved (dashed) variables.
- **Edges:** arrows that encode the presence or absence of a causal effect.
 - Arrow present = a direct causal effect: $Y_i(d) \neq Y_i(d')$ for some i and d .
 - Lack of an arrow = no causal effect: $Y_i(d) = Y_i(d')$ for all i and d .
 - Missing variables = no other common causes of any variables.
- **Directed:** each arrow implies a direction (causal ordering).
- **Acyclic:** no cycle: a variable cannot cause itself

DAG Terminologies



- **Path:** a sequence of edges that connect two nodes.
 - A **directed** or **causal** path is all in the same causal direction.
 - Non-causal path example: $D \leftarrow X \rightarrow Y$
- **Descendants:** nodes on a directed path away from some other node.
 - M is a descendant of D and X .
 - Ancestors is the reverse: X is an ancestor of M .
- **Parents:** immediate causes of a node.
 - D is the parent of Y and M .
 - **Children** are the reverse: M is a child of D .

DAGs to Distributions



$$Y = f_y(D, U, \varepsilon_y)$$

$$D = f_d(Z, X, \varepsilon_d)$$

$$X = f_x(U, \varepsilon_x)$$

$$Z = f_z(X, \varepsilon_z)$$

- Causal DAGs equivalent to **nonparametric structural equation models**
- NPSEM have a **causal interpretation**, but are completely flexible.
 - No specification of a functional form or interactions, etc.
 - More standard linear SEM is a special case.
- Causal DAGs imply the following factorization:

$$\mathbb{P}(X_1, X_2, \dots, X_J) = \prod_{j=1}^J \mathbb{P}(X_j \mid \text{pa}(X_j)) \quad \text{where } \text{pa}(X_j) = \text{parents of } X_j$$

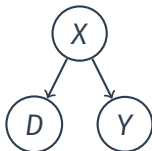
$$\mathbb{P}(D, U, X, Y, Z) = \mathbb{P}(Y \mid D, U) \mathbb{P}(D \mid X, Z) \mathbb{P}(Z \mid X) \mathbb{P}(X \mid U) \mathbb{P}(U)$$

D-Separation

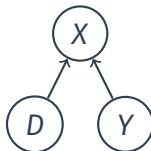
- Can we determine conditional independence from our causal DAG?
- Yes! To verify that $A \perp\!\!\!\perp B \mid C$ where each is a set of nodes:
 1. Find all paths from any vertex in A to any vertex in B .
 2. Check if each path is **blocked**.
 3. If all paths are blocked, then A is **d-separated** from B by C .
- A path is **blocked** conditional on C if:
 1. C includes a non-collider on that path **OR**
 2. Path includes a collider not in C and no descendant of any collider is in C .
- If A and B are d-separated, then we have $A \perp\!\!\!\perp B \mid C \rightsquigarrow$ if not, then d-connected.

Common Structures

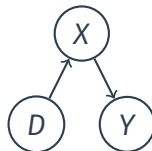
Confounder



Collider

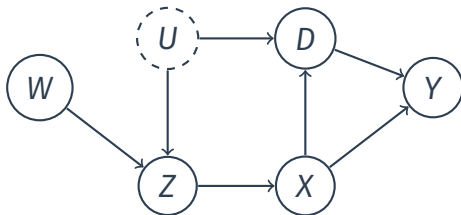


Mediator



- **Confounder** (fork): Common cause of two variables.
 - D and Y unconditionally dependent, conditionally independent.
- **Collider** (inverted fork): common descendent of two variables.
 - D and Y unconditionally independent, conditionally dependent.
 - X “blocks” the relationship between them when not conditioned on.
 - Example: D (education); Y (experience); X (hired).
 - Now, $X=1$ with low D must mean high Y , vice versa.
- **Mediator** (chain): variable on the path from one variable to another.
 - D and Y unconditionally dependent.

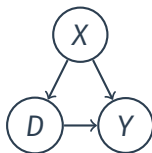
D-Separation Example



- Are W and Y marginally independent of each other?
 - Blocked: $W \rightarrow Z \leftarrow U \rightarrow D \rightarrow Y$ $W \rightarrow Z \leftarrow U \rightarrow D \leftarrow X \rightarrow Y$
 - Unblocked: $W \rightarrow Z \rightarrow X \rightarrow Y$ $W \rightarrow Z \rightarrow X \rightarrow D \rightarrow Y$
- Which variables should we condition on to make W and Y conditionally independent (d-separated)?
 - Block the unblocked paths without unblocking the blocked paths.
 - Conditioning on X would do this.
 - Conditioning on D and/or Z would unblock some of the blocked paths because they are colliders.

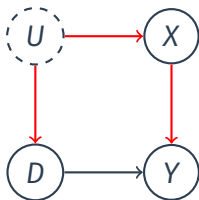
Backdoor Paths and Blocking Paths

- **Backdoor paths:** non-causal path from D to Y .
 - Would remain if we removed any arrows pointing out of D .
- Backdoor paths between D and $Y \rightsquigarrow$ common causes of D and Y :



- Here: backdoor path $D \leftarrow X \rightarrow Y$

Other Types of Confounding



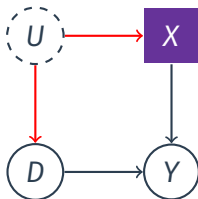
- D : enrolling in a job training program.
- Y : getting a job.
- U : being motivated.
- X : number of job applications sent out.
- Big assumption here $\rightsquigarrow$ no arrow linking U to Y

Backdoor Criterion

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \mid \mathbf{X}_i$$

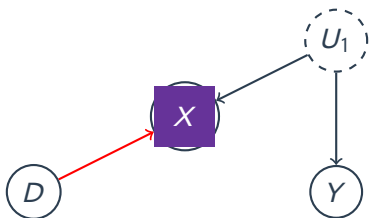
- Can we use a DAG to evaluate no unmeasured confounders?
- Holds if the **backdoor criterion** on a causal DAG is met:
 1. No vertex/node in $\mathbf{X}$ is a descend of D (**no post-treatment bias**), and
 2. $\mathbf{X}$ blocks all backdoor paths from D to Y .
- The backdoor criterion is pretty powerful. Tells us:
 1. If there is confounding given this DAG,
 2. If it is possible to remove the confounding, and
 3. What variables to condition on to eliminate the confounding.

Other Types of Confounding *Redux*



- D : enrolling in a job training program.
- Y : landing a job!
- U : being motivated.
- X : number of job applications sent out.
- Big assumption here $\rightsquigarrow$ no arrow linking U to Y
- Conditioning on X blocks all backdoor paths.

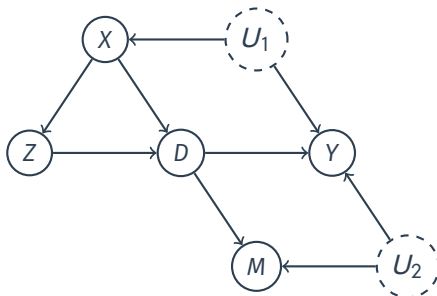
Why Not Condition on Descendants?



- No causal or statistical relationship between D and Y
- Conditioning on post-treatment variables opens up non-causal paths
 - $\rightsquigarrow$ Statistical relationship between D and Y conditional on X
 - But still no causal relationship $\rightsquigarrow$ selection bias.

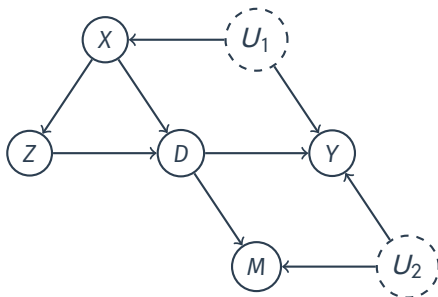
Usage Example of DAG

Assume that you've come out with a DAG based on your expertise:



- Suppose you want to identify a causal effect of D on Y .
 - In a nutshell, you want to block all the paths that yield statistical associations between D and Y .
 - Thus, you want to find a set of nodes S such that once we condition on S
 - **no unmeasured confounding** holds and
 - any descendant of D is not in $S \rightarrow$ no post-treatment bias.
- $\rightsquigarrow$ Use **backdoor criterion**!

Back Door Criterion Example



1. List all of the **backdoor paths** between D and Y .
2. List all the possible set of nodes $\mathbf{S}$ that you can condition on.
3. List all the $\mathbf{S}$ such that **blocks** all the backdoor paths.
4. Among those $\mathbf{S}$, drop the sets which include a descend of D .

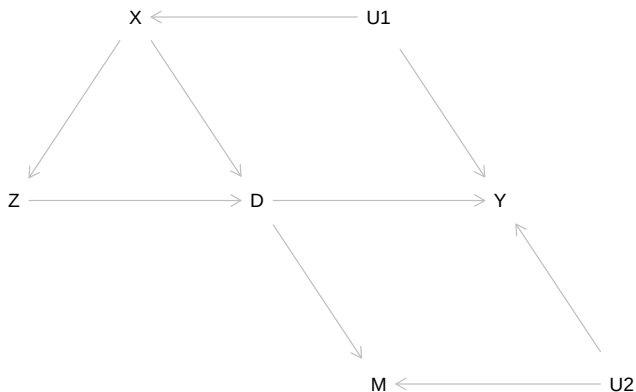
R Package: dagitty

- DAGitty: www.dagitty.net

```
1 > library(dagitty)
2 > g <- dagitty('dag {
3     X [pos="1,-1.5"]
4     Y [pos="4,0"]
5     Z [pos="0,0"]
6     M [pos="3,1.5"]
7     D [pos="2,0"]
8     U1 [pos="3,-1.5"]
9     U2 [pos="5,1.5"]
10    X -> Z -> D -> Y
11    X -> D -> M
12    M <- U2 -> Y
13    X <- U1 -> Y
14  }')
15 > latents(g) <- c("U1", "U2")
```

R Package: dagitty

```
1 > plot(g) # Visualize the DAG
```



R Package: dagitty

- Access parent/ancestor nodes:

```
1 > parents(g, "D") ## [1] "X" "Z"  
2 [1] "X" "Z"  
3  
4 > ancestors(g, "D") ## [1] "D" "Z" "X" "U1"  
5 [1] "D" "Z" "X" "U1"
```

- Or children/descendent nodes:

```
1 > children(g, "D") ## [1] "M" "Y"  
2 [1] "M" "Y"  
3  
4 > descendants(g, "D") ## [1] "D" "Y" "M"  
5 [1] "D" "Y" "M"
```

R Package: dagitty

- Identify paths using `paths()`:

```
1 > paths(g, "D", "Y")$paths
2 [1] "D -> M <- U2 -> Y"      "D -> Y"      "D <- X <- U1 -> Y"
3 [4] "D <- Z <- X <- U1 -> Y"
```

- Extract causal path(s) by setting `directed = T`:

```
1 > paths(g, "D", "Y", directed = TRUE)$paths # only causal path(s)
2 [1] "D -> Y"
```

- Check whether two nodes are d-separated using `dseparated()`:

```
1 > dseparated(g, "Z", "D", c("X")) # because of Z -> D
2 [1] FALSE
3
4 > dseparated(g, "Z", "M", c("D"))
5 [1] TRUE
```

R Package: dagitty

- Access the set of nodes **S** to condition on for **no unmeasured confounding** to hold:

```
1 > adjustmentSets(g, "D", "Y", type="minimal")
2 { X }
```

- Caveat: adjustmentSets may include unobserved variables which we cannot actually condition on.

```
1 > S = adjustmentSets(g, "D", "Y", type="all")
2
3 > S[!grepl("U1|U2", S)]
4 { X }      # X typically preferred for simplicity and max statistical power, unless
5 { X, Z }   # Z is a known strong confounder and you have sufficient data.
```

- Note that this implements a slightly more general criterion (sometimes it may contain descendants)

R Package: dagitty

- Full list of adjustment sets:

```
1 # S = adjustmentSets(g, "D", "Y", type="all")
2 > S
3 { U1 }
4 { U1, U2 }
5 { M, U1, U2 }
6 { X }
7 { U1, X }
8 { U2, X }
9 { M, U2, X }
10 { U1, U2, X }
11 { M, U1, U2, X }
12 { U1, Z }
13 { U1, U2, Z }
14 { M, U1, U2, Z }
15 { X, Z }
16 { U1, X, Z }
17 { U2, X, Z }
18 { M, U2, X, Z }
19 { U1, U2, X, Z }
20 { M, U1, U2, X, Z }
```

DAGitty (<https://www.dagitty.net/>)

Variable

E

☒ exposure
☐ outcome
☐ adjusted
☐ selected
☐ unobserved

delete rename

View mode

☒ normal
☐ moral graph
☐ correlation graph
☐ equivalence class

Effect analysis

☐ atomic direct effects

Diagram style

☒ classic
☐ SEM-like

Coloring

☒ causal paths
☒ biasing paths
☒ ancestral structure

Legend

☒ exposure

☒ outcome

☐ ancestor of exposure

☐ ancestor of outcome

☐ ancestor of exposure and outcome

☐ adjusted variable

☐ unobserved (latent)

☐ other variable

☒ causal path

☐ biasing path

Model | Examples | How to ... | Layout | Help

```
graph TD; A[A] --> Z((Z)); B[B] --> Z; E((E)) --> D[D]; A --> D; B --> A;
```

Causal effect identification

Adjustment (total effect)

Exposure: E
Outcome: D
Selected: A
Adjusted: Z
Correctly adjusted.

Testable implications

The model implies the following conditional independences:

- $A \perp B$
- $A \perp D \mid E$
- $B \perp E$
- $D \perp Z \mid A, B$
- $D \perp Z \mid B, E$
- $E \perp Z \mid A$

Model code

dag {
A
[selected,pos="-2.200,-1.520"]
}
B [pos="1.400,-1.460"]
D [outcome,pos="1.400,1.621"]
E [exposure,pos="-2.200,1.597"]
Z
[adjusted,pos="-0.300,-0.082"]
}

Summary

exposure(s) **E**
outcome(s) **D**
covariates **3**
causal paths **1**

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Onto the presentations & discussions!

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

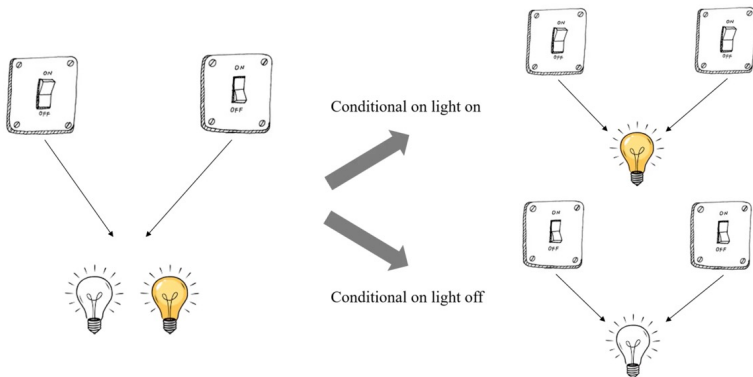
<https://j1yoo.github.io/>



Appendix

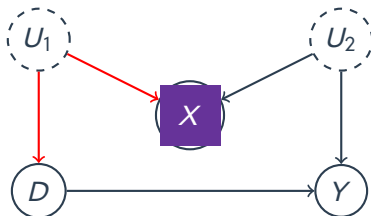
Conditioning on Collider

- Two independent switches that turn the light on/off:



Sources: Jiyong Park (UNC at Greensboro), Korea Summer Session on Causal Inference 2021

M-Bias



- Not all backdoor paths induce confounding:
- No conditioning: backdoor path blocked by the collider X .
- If we control for $X \rightsquigarrow$ this “opens up” the path and induces confounding!
 - Sometimes referred to as **M-bias** or **collider bias**.
- Controversial because of differing views on what to control for:
 - Rubin thinks that M-bias is a “mathematical curiosity” and we should control for all pretreatment variables.
 - Pearl and others think M-bias is a real threat.